Retail Sentiment Intelligence

Real-Time Social Media Mining and Trust-Aware Sentiment Analysis Using Large Language Models for Retail Feedback Optimization

BITS ZG628T · Dissertation · Post Mid-Semester Presentation

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Agenda

Focus of today — post mid-semester work only

1. **Since Mid-Sem — What's New**
   Post-midsem workstream map
2. **Review & Validate (HITL)**
   Human-in-the-loop correction workflow
3. **Smart Reply — Triple Draft**
   GPT-4o + Mistral 7B + Smart Composer
4. **Smart Reply — Prompt & Few-Shot**
   Prompt template + past validated replies
5. **Smart Reply — Worked Example**
   One complaint → three drafts
6. **Learning Loop**
   Corrections + posted replies → retraining
7. **Post Explorer**
   Multi-facet search across analysed posts
8. **Post Lifecycle (Kanban)**
   Triage → Acknowledged → In-Progress → Resolved
9. **Insights & Competitor Analysis**
   Strategic weekly summaries + cross-brand
10. **Notification Centre**
    In-app + email + Slack, group-routed
11. **ModernBERT — Final Results**
    Macro-F1 0.6272 → 0.7642 (+13.7 pts)
12. **Vision — Multi-Pass Payoff**
    Hallucination 50% → 0%, extraction 25% → 75%
13. **Trust-Score Evaluation**
    15% flagged; 12 of 15 confirmed by annotator
14. **Live Demo**
    Dashboard walkthrough (screenshots)
15. **Conclusions & Future Work**
    RQ1–RQ4 outcomes + realistic roadmap

Since Mid-Sem — What's New

Mid-sem shipped the pipeline; post-midsem made it usable for analysts

DELIVERED AT MID-SEM

- 6-layer offline-first pipeline (25 subreddits)
- ModernBERT fine-tuning up to Stage 2 (F1 0.7285)
- Multi-pass vision with 0% hallucination on pilot
- Trust-score design + weighted formula
- Brand-Health + Alert Feed dashboard pages
- Group-routed email / Slack notifications

ADDED POST-MIDSEM (this deck)

- Review & Validate — human-in-the-loop UI + backend
- Smart Reply Composer — triple-draft (GPT-4o + Mistral + Smart)
- Learning-loop store — corrections & posted replies
- Post Explorer — multi-facet search
- Post Lifecycle — Kanban with 2-step Resolve
- Insights & Competitor Analysis page
- In-app Notification Centre mirroring the digest
- ModernBERT Stage 3 (final) — 0.7285 → 0.7642 (5-fold OOF)
- Trust-score end-to-end evaluation (n = 200)

THE FRAMING QUESTION

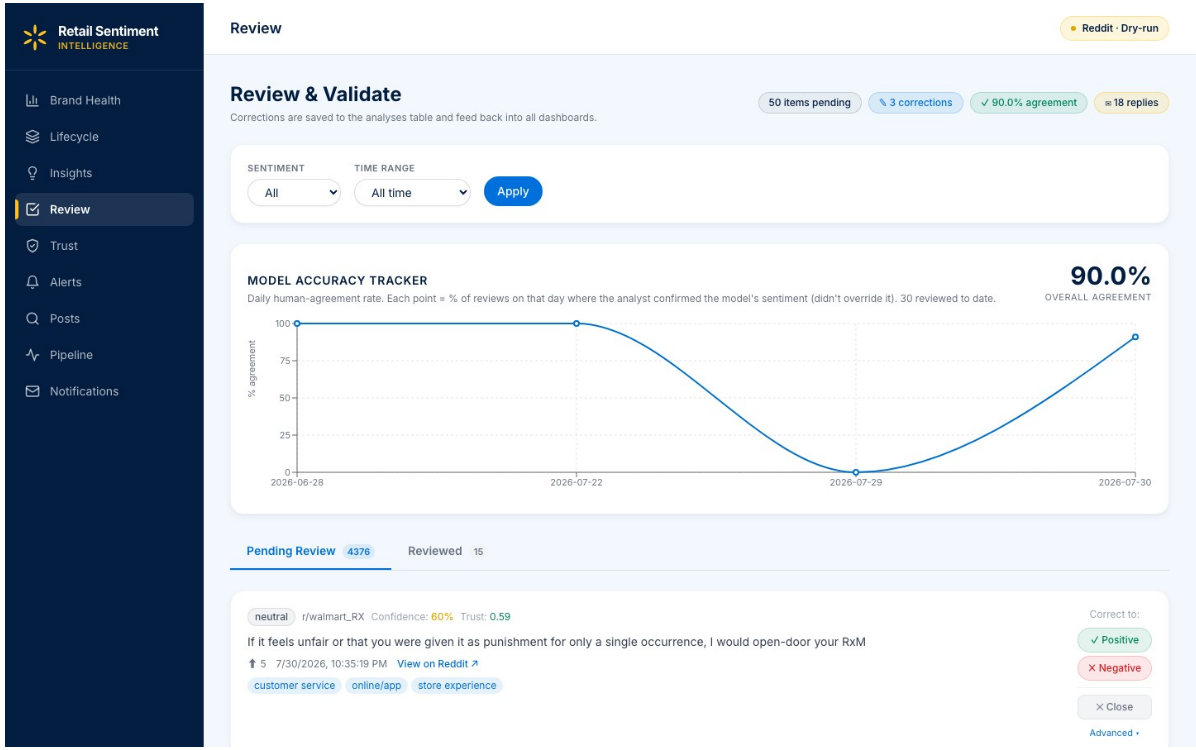
“ Mid-sem answered can the pipeline produce trust-weighted sentiment? Post-midsem answers can an analyst act on it, correct it, and feed the corrections back into the model?

Review & Validate — Human-in-the-Loop

Every prediction can be corrected; every correction is logged

WORKFLOW

- Queue sorted by priority (P1 first) · trust × confidence
- Analyst can override sentiment and aspect tags in one click
- Corrections written to feedback table with analyst + timestamp
- Generate Drafts → three reply options (GPT-4o + Mistral + Smart)
- Analyst picks A, B or C, edits inline, and posts to Reddit
- Posted replies also captured → future few-shot pool



WHY IT MATTERS

Turns a passive dashboard into a training signal — corrections harvested here drive the ModernBERT re-training loop and reply few-shot pool.

NUMBERS TODAY (live capture)

18 posted replies logged · 60 total feedback rows · ≈90 % analyst – model agreement on the last 30 reviews.

Smart Reply Composer — Triple Draft

One prompt, three generators; analyst picks the best draft

Draft A	Draft B	Draft C
GPT-4o	**Mistral 7B-Instruct**	**Smart Composer**
Walmart LLM Gateway	Local Ollama (:11434)	Deterministic template
Strong reasoning, safest tone Guardrails via Walmart proxy ~$0.0002 / reply Fallback: direct OpenAI	Open-weights, free Runs fully offline ~15 s warm latency Strong at retail jargon	No LLM, always available Zero latency, zero cost Content-aware phrase pools Safety-net fallback

Fallback logic — no draft is ever missing

- GPT-4o unavailable (no gateway key / consumer-ID / network) → Smart Composer draft in slot A
- Mistral unavailable (Ollama not running) → Smart Composer draft in slot B
- Smart Composer always produces slot C — guaranteed reply
- UI labels each card with the actual model used, plus an offline-fallback badge when applicable

Smart Reply — Prompt Design & Few-Shot

Same prompt fed to GPT and Mistral; Smart Composer skips the LLM step

Prompt template (built in _build_reply_prompt)

```
You are a senior Walmart customer-care analyst replying on Reddit.
Write ONE reply to the customer below. Keep it 2-4 sentences,
empathetic, specific to their complaint, no corporate jargon,
no hashtags, no emojis. Do NOT promise refunds you can't verify;
invite them to DM order details if action is needed.
Sign off as a real person, not a brand.

Example customer post: {past_post_1}
Example analyst reply: {past_reply_1}
 ... (three example pairs total)

Subreddit: r/{subreddit}
Customer ({author}) complaint about: {aspect_1}, {aspect_2}
Customer post: {post_title + post_body, ≤ 1200 chars}

Reply:
```

PROMPT ANATOMY

① **SYSTEM ROLE**
who the model is + tone rules

② **FEW-SHOT**
3 past validated reply pairs
from the feedback table

③ **CONTEXT**
subreddit · author · top-2 aspects

④ **POST**
title + body, truncated to 1 200 chars

⑤ **TRIGGER**
"Reply:" cues the model to generate

Few-shot source — feedback table

- Top-3 past validated replies for the same aspect
- Written by human analysts, already posted to Reddit
- Refreshed on every Generate click → adapts over time

Style guardrails baked into the prompt

- 2 – 4 sentences, empathetic, specific
- No corporate jargon, no hashtags, no emojis
- No unverifiable refund promises → DM to continue
- Sign off as a person, not a brand

Smart Reply — Worked Example (all three drafts real, live-captured)

raw_posts.id = reddit_1u2bgdw · r/samsclub · gateway ✓ · Ollama ✓ · Smart Composer ✓

CUSTOMER COMPLAINT · r/samsclub

Why do you guys sell whole pizzas made hours ago to customers?

"The very few times I've gotten a whole pie, it'll be stuff premade and left in the hot case for like an hour 30mins before it's in my hand. How can I tell? They put a sticker with the date and the pizza looks and taste hours old. Pizza is meant to be made to order and waited for."

ModernBERT: negative (conf 0.9999997) · aspects: customer service · product quality · store experience · trust 0.66

🔗 Open original on Reddit

A · GPT-4o (Walmart Gateway)

Hi u/there, I'm sorry to hear about your experience with the pizzas. I understand how disappointing it must be to get something that feels less fresh than expected. While the hot case is meant to provide convenience, I agree that quality should never be sacrificed. If you'd like, feel free to DM me with the details of your store location so we can look into this and address it properly. – [Your Name]

B · Mistral 7B (local Ollama)

Hi u/there, I appreciate your concern about the quality of our prepared pizzas. It's important to us that our customers receive fresh and delicious food. The stickers with dates are there to help us manage the rotation of our products, ensuring they are consumed before they become stale. If you ever encounter a pizza that doesn't meet your expectations, please feel free to DM me the details of your order so I can look into it further. Thank you for taking the time to share your feedback.

C · Smart Composer (no-LLM)

Hi u/there — thanks for flagging this. completely understand the frustration around your order. Drop us a private message with the order # or store info and we'll take it from here. — Walmart Care

WHAT HAPPENS NEXT

One prompt fed to all three engines. Analyst picks the best draft, edits inline, clicks Save & Open Reddit → the posted reply is written to feedback and becomes the next post's top few-shot example.

Learning Loop — Feedback → Retraining

Every human action becomes a training signal — hot loop (live) + warm loop (monthly)

Label / aspect correction	Trust-score override	Posted reply	Lifecycle transition
Review & Validate	Review & Validate	Smart Reply Composer	Kanban board

feedback table — one row per human action, JSON payload, partition_key = analyst_id

HOT LOOP · in-context, live

- Every Generate Drafts click queries feedback
- SELECT ... WHERE kind = 'auto_reply_posted'
- ORDER BY created_at DESC LIMIT 5 → top 3 pairs
- Injected into the prompt as few-shot examples
- Effect visible on the next reply — no training

WARM LOOP · supervised, monthly

- Export corrections since last train date
- Append to Walmart-200 → Walmart-N (augmented)
- Rerun ModernBERT Stage-3 with 5-fold OOF CV
- New checkpoint wins only if OOF F1 improves
- Symlink flipped in config/models.yaml

Two failure modes eliminated by design

(1) hot loop uses in-context learning — no retrain, no catastrophic forgetting. (2) warm loop uses 5-fold OOF CV — no leakage between train and eval sets.

Post Explorer — Multi-Facet Search

Find the exact 20 posts an analyst needs to look at today

FACETS

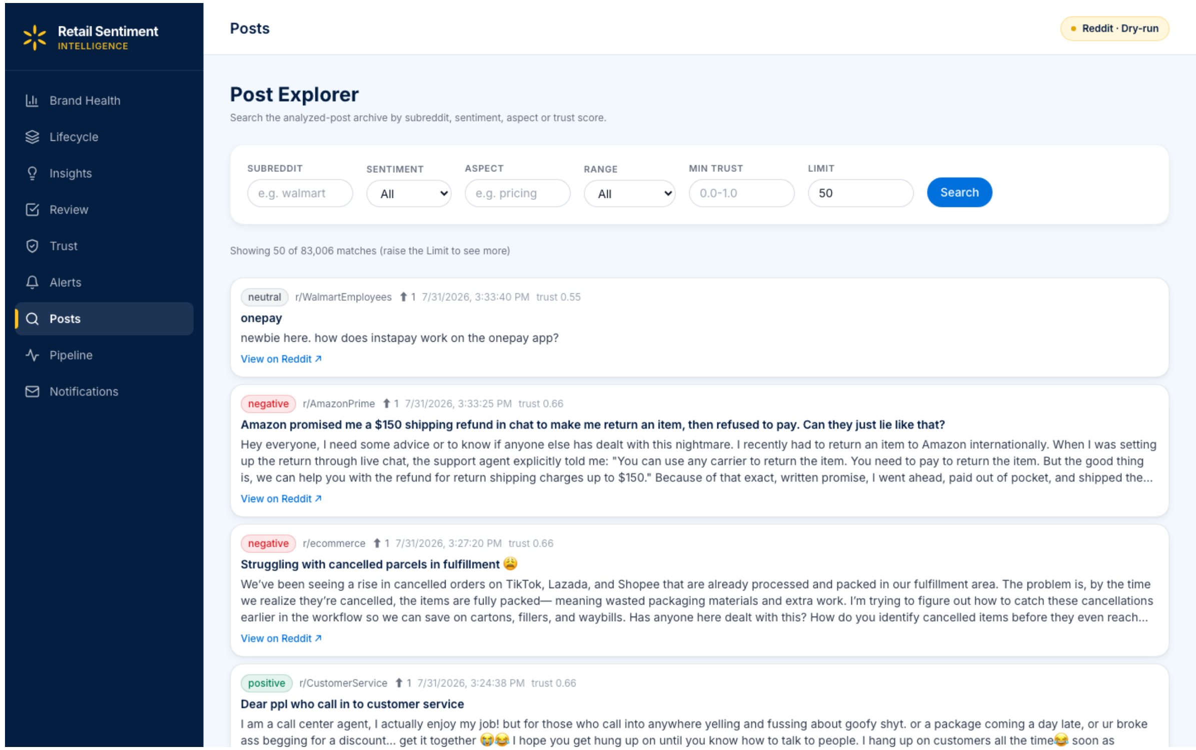
- Sentiment · neg / neu / pos
- Confidence slider (0.50 – 1.00)
- Trust-score slider (0.00 – 1.00)
- Subreddit — multi-select from 25 tracked
- Aspect — 8-item retail taxonomy
- Date range — today / week / month / custom
- Full-text search — title + body

PER-POST CARD

Title + body excerpt · sentiment badge with confidence % · trust tier (H / M / L) · aspect chips · subreddit + timestamp · actions: Review · Add to Lifecycle · View Details.

CATALOG TODAY (live capture)

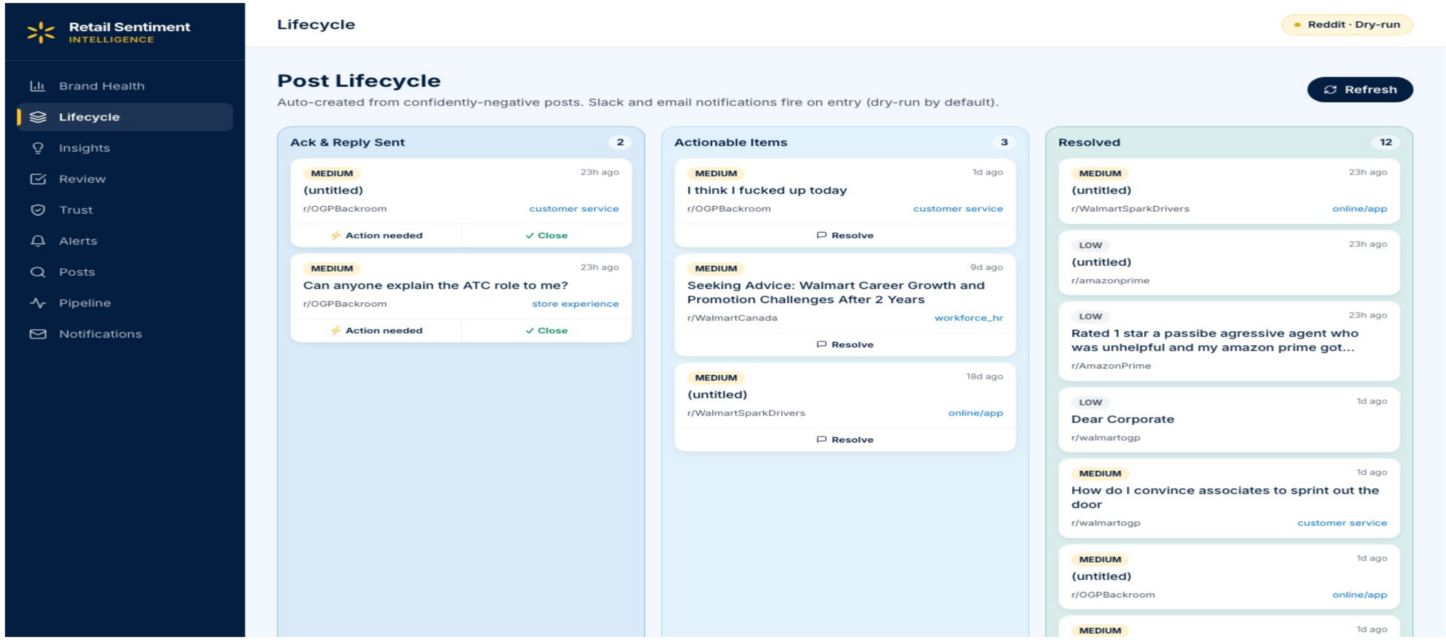
83 006 analysed posts · 25 subreddits · 8 aspects · results paginated at 50 / query.



Post Lifecycle — Kanban Workflow

Track a complaint from Triage to Resolved with SLA visibility

TRIAGED	ACKNOWLEDGED	IN PROGRESS	RESOLVED
New P1 / P2 posts land here	Assigned to analyst	Reply being drafted / posted	Reply posted or no-action closed



RESOLVE MODAL — 2-step flow

- Step 1 · save action note + optional LLM reply
- (a) Save & open Reddit → reply copied to clipboard
- (b) OR Resolve (no-reply) → close without posting
- Step 2 · paste on Reddit → Mark Resolved
- All transitions timestamped → SLA analytics

Insights & Competitor Analysis

Strategic view — issue rankings, cross-brand pulse, LLM summaries

Priority-Negatives

Top issues ranked by
volume × severity × recency
by aspect

Competitor Pulse

Walmart vs Costco,
Target, Amazon on shared aspects

Weekly LLM Summaries

Natural-language weekly
digest + action items
+ emerging topics

Aspect Drilldown

8-aspect taxonomy
per-aspect sentiment trend
+ representative posts

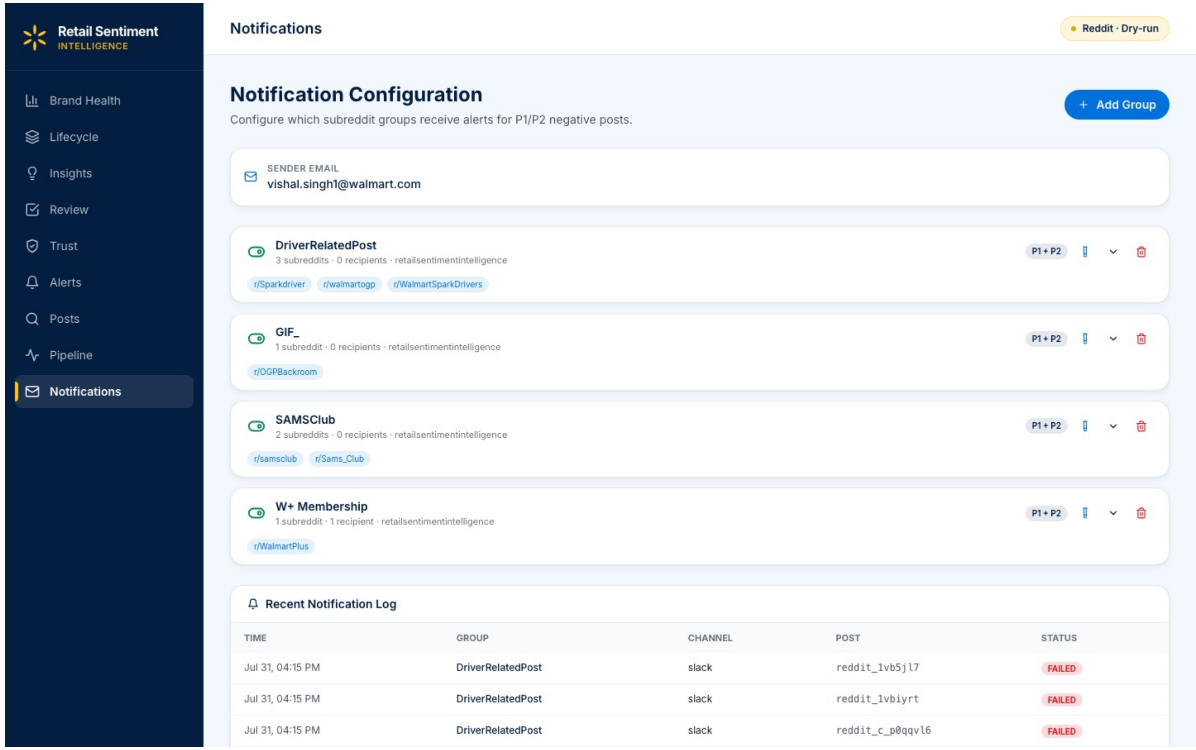
Weekly summaries powered by GPT-4o via Walmart Gateway · see Live Demo grid on slide 15 for the actual page.

Notification Centre

In-app mirror of the group-routed email / Slack digest

WHAT IT DOES

- Every P1 / P2 email or Slack alert is mirrored in the app
- Notifications grouped by subreddit ↔ business team
- Read / unread state persisted per analyst
- Deep-links open the post in Review & Validate
- Powered by the same alerts rows that drive email/Slack



PRIORITY THRESHOLDS

P1 — trust ≥ 0.70 and confidence ≥ 0.80
P2 — trust ≥ 0.50 and confidence ≥ 0.60
Source: config/models.yaml + dispatcher.py

GROUPS TODAY (live capture)

4 routing groups configured · DriverRelatedPost, GIF_, SAMSClub, W+ Membership · delivery status tracked per row in notification_log.

ModernBERT — Final Training Results

3-stage curriculum, 5-fold out-of-fold CV, n = 200

Overall Macro-F1	Long posts (≥ 512 tok, n = 7)	Uplift over baseline
0.6272 → 0.7642	5 / 7 → 7 / 7 correct	+13.7 pts

3-stage curriculum

Stage	Data	Macro-F1	Δ vs baseline
Baseline (Twitter-RoBERTa)	off-the-shelf	0.6272	—
Stage 1	TweetEval sentiment (~45 k tweets)	0.6810	+5.4
Stage 2	GoEmotions-3class Reddit (~54 k, pseudo)	0.7285	+10.1
Stage 3 (final)	Walmart-200 hand-labelled	0.7642	+13.7

Per-length-bucket accuracy (5-fold OOF predictions)

Bucket (token count)	Baseline correct	ModernBERT correct	Recovered
Short-to-medium (< 512, n = 193)	138 / 193 (72 %)	159 / 193 (82 %)	+21
Long (≥ 512, n = 7, all negative)	5 / 7	7 / 7	+2

Vision Pipeline — Multi-Pass Payoff

Same model (Gemma 3 4B), same hardware, smarter calling strategy

PASS 1

STRUCTURE

What TYPE of image?
screenshot / photo /
receipt / app / meme

PASS 2

TILE

Split into 2–4 crops
→ 2–4× effective
resolution, no resize

PASS 3

EXTRACT

Per tile: read ALL
text verbatim
(quotes, prices, UI)

PASS 4

MERGE

Text-only LLM call
— image REMOVED —
cannot invent visuals

Evaluation on 32 retail screenshots

Metric	Single-pass	Multi-pass	Change
Hallucination rate	50 %	0 %	eliminated
Text-extraction success	25 %	75 %	3× better
Retail-signal recall	40 %	81 %	2× better
Fabricated claims	many	0	eliminated
Latency / image (warm)	~5 s	~15 s	3× (accepted)

Trust-Score — End-to-End Evaluation

On the 200-post gold set, does the trust filter catch bad posts?

Low-trust share (T < 0.4)	Human-agreed (annotator cross-check)	Agreement rate
15 %	**12 / 15**	**80 %**

Weighted formula (decomposition shown inline in dashboard)

$$\text{trust\_score} = 0.4 \times \text{metadata} + 0.3 \times \text{dedup} + 0.3 \times \text{llm\_credibility}$$

DESIGN PRINCIPLE — flag, don't drop

Low-trust posts are surfaced with an explanatory chip (promotional / duplicate / thin-metadata) and can be overridden by an analyst in one click — the override is captured for the learning loop.

WHY IT'S DEFENSIBLE

- Every constant has a plain-English rationale in config/models.yaml
- Three named components → analyst can see WHY a post is low-trust
- 12 / 15 confirmed by human annotator on the cross-check
- LLM component only invoked in ambiguous zone (cost-controlled)

Evaluation Summary — Post-Midsem Numbers

All results reported against the same 200-post retail-Reddit gold set

Area	Metric	Value	Notes
Sentiment	Macro-F1 (final)	0.7642	+13.7 pts vs RoBERTa baseline
Sentiment	Long-bucket accuracy (≥ 512 tokens, n=7)	7 / 7 correct	Baseline 5 / 7 · all negative-class
Sentiment	Cross-validation	5-fold OOF	No test-set leakage
Vision	Hallucination rate	0 %	From 50 % on single-pass
Vision	Text extraction success	75 %	From 25 % on single-pass
Vision	Retail-signal recall	81 %	From 40 % on single-pass
Trust	Low-trust share (T < 0.4)	15 %	Of the 200-post set
Trust	Annotator agreement on low-trust	12 / 15	80 %
Ops	Subreddits tracked	25	Six community groups
Ops	Scheduler cadence	6 h + manual	asyncio lifespan task
Ops	Dashboard pages shipped	8	Brand Health · Alerts · Explorer · Review · Lifecycle · Insights · Pipeline · Notifications

Principal Contributions

What this dissertation delivered — end to end

C1

Offline-first RSI pipeline

Ingestion → trust → sentiment → aspects → vision → aggregation → alerts → dashboard. Zero API cost, modular, deployable to Azure.

C2

Fine-tuned ModernBERT (3-stage curriculum)

Macro-F1 0.6272 → 0.7642 overall; recovers all long posts (≥ 512 tok, n = 7): RoBERTa 5/7 vs ModernBERT 7/7 correct. 5-fold OOF cross-validation, offline training.

C3

Multi-pass vision technique on Gemma 3 4B

Hallucination 50 % → 0 %, extraction 25 % → 75 %. Adapts techniques from 5 recent VLM papers without vendor-blocked models.

C4

Interpretable trust score + admission gate

trust × confidence gate flags rather than drops low-credibility posts. Every constant traceable to English rationale in config.

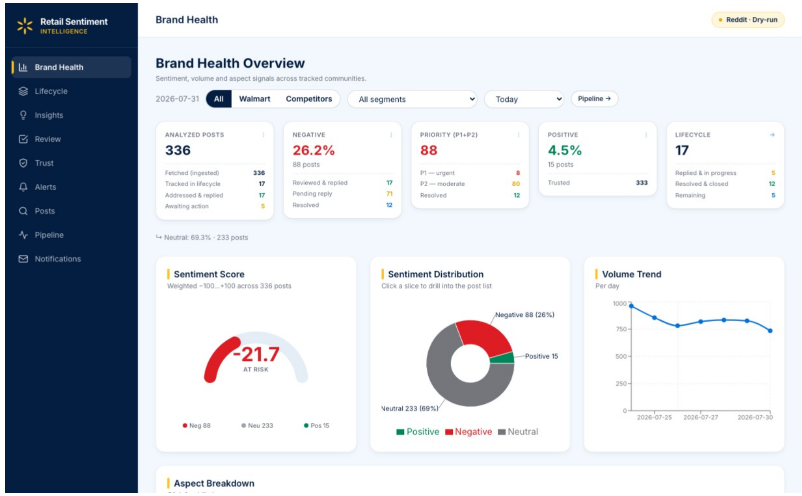
C5

HITL learning-loop dashboard

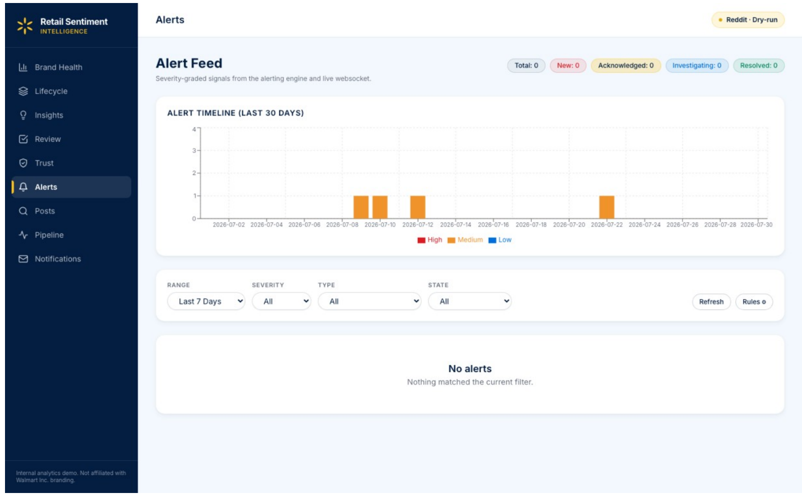
Review & Validate, Lifecycle Kanban, Insights, Notification centre. Corrections & posted replies feed few-shot reply generation and future retraining.

Live Demo — Dashboard Walkthrough

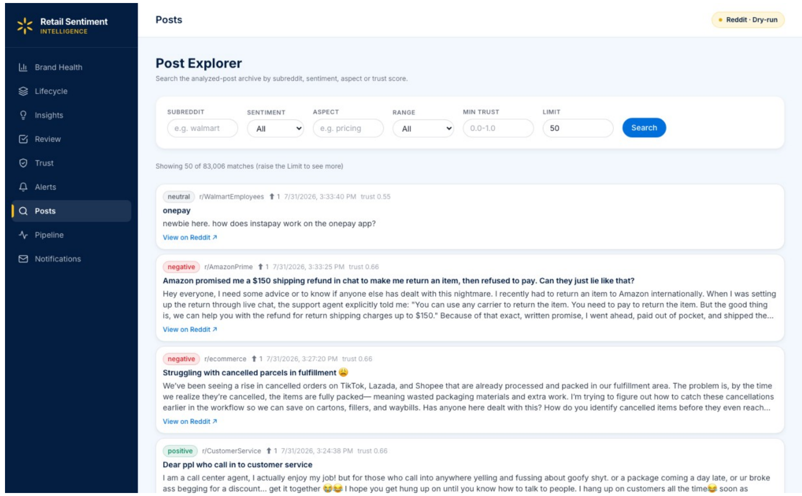
One post → alert → review → reply → lifecycle → resolved



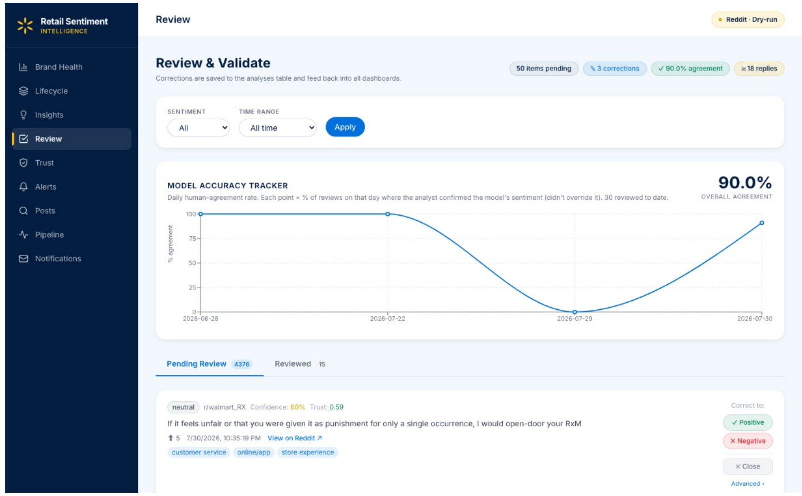
Brand Health



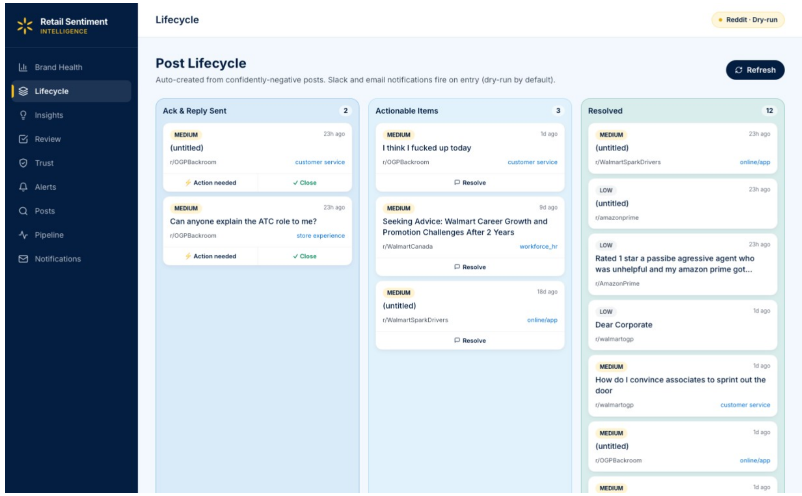
Alert Feed



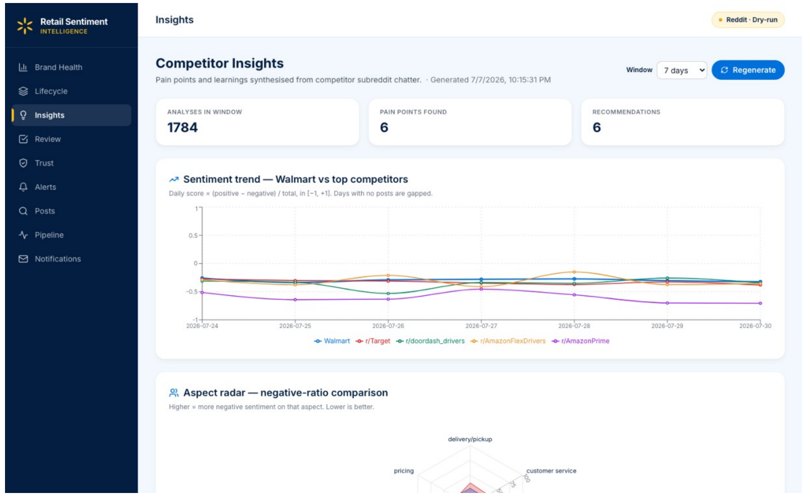
Post Explorer



Review & Validate



Lifecycle Kanban



Insights

Conclusions — What Worked, What's Still Open

Honest read on 5 months of post-midsem work

WHAT WORKED

- 3-stage ModernBERT curriculum landed the sentiment win we set out for (Macro-F1 0.62 → 0.76 on 5-fold OOF).
- Long-post recovery came for free once we switched off truncation (≥1024-tok context): 5/7 → 7/7 correct on the ≥ 512-tok bucket.
- Removing the image on the final vision merge step ended the 50 % hallucination problem — the mechanism, not just the number, makes it credible.
- Trust score as flag-not-drop matched the human annotator on 12 / 15 low-trust posts. Analysts trust it because they can override it.
- HITL feedback table quietly turned into the most useful piece of infrastructure — drives few-shot on every click today, retraining tomorrow.

WHAT'S STILL OPEN (honest)

- 200-post gold set is small. Numbers are OOF-CV, but a bigger blind held-out set is needed before I'd claim generalisation.
- Vision eval is 32 images — enough to sanity-check the mechanism, not enough to claim production-grade quality.
- Long bucket has only 7 posts and they are all negative-class — 7/7 correct is evidence the truncation ceiling is gone, not proof of long-text mastery.
- Reply drafts still need an analyst edit — we track edit-distance but haven't shown it dropping consistently yet.
- Everything runs on one Mac. Multi-analyst concurrency and retraining automation aren't done — in the roadmap on next slide.

THE TAKEAWAY

Mechanism-level wins are what carry the argument; sample-size caveats are what the next horizon of work has to close.

Future Work — 4-Horizon Roadmap

From today's HITL to a Walmart-trained, mostly-autonomous replier

Now → 1 month

HITL: 100 %

Grow the gold set

Grow the labelled gold set to ~500 posts using the HITL feedback the team is already producing; rerun 5-fold OOF to check the F1 delta.

1 → 3 months

HITL: 100 %

Automate + go bilingual

Wire the monthly ModernBERT retrain into a Cron / Airflow job with an OOF-F1 promotion gate. Add multi-language support (start with Hindi + English) — taxonomy validated with native-speaker analysts.

3 → 6 months

HITL: ~80 %

Walmart-trained model + shared DB

Fine-tune Mistral 7B on our own posted-reply corpus → a Walmart-trained replier that drops the paid GPT-4o dependency, runs fully in-house, and can be self-hosted. Move the feedback table to a shared managed database so multiple analysts can work concurrently.

6 → 12 months

HITL: ~30 % → eventually 0 %

Confidence-gated auto-reply

Introduce a confidence gate on the reply generator: top-confidence drafts are queued for auto-send and the analyst only reviews the mid- and low-confidence tail. As agreement stays high, gradually raise the auto-send threshold — HITL winds down to sampling / audit only.

GUIDING PRINCIPLE

HITL is the ladder, not the ceiling — every stage produces the training signal that lets the next stage automate more. Each horizon is admissible only when the previous one is measurably successful.

Source Code & Deliverables

Repository

https://gecgithub01.walmart.com/v0s01jh/Retail_Sentiment_Intelligence

Deliverable	Path in repo	Contents
Final Report (PDF)	docs/Sem_4/final/FINAL_REPORT_VishalSingh_2020AA05641.pdf	10 chapters + appendices + refs
Final Report (LaTeX)	docs/Sem_4/final/latex/	Reproducible xelatex source
Pipeline core	src/pipeline.py, src/ingestion/, src/analysis/	6-layer async pipeline
Sentiment training	scripts/train_modernbert_sentiment.py	3-stage curriculum runner
Evaluation notebook	evaluation/trust_score_walmart200.ipynb	Reproducible 5-fold OOF eval
Dashboard	frontend/ (React + Vite + Tailwind)	8 pages, live via WebSocket
Reproduction	Appendix B of the report · start.sh	Clone → conda → start

REPRODUCE IN 3 STEPS

git clone <repo> → conda create -n rsi python=3.11 && pip install -r requirements.txt → ./start.sh

Thank You

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Questions ?

Repository

gecgithub01.walmart.com/v0s01jh/Retail_Sentiment_Intelligence

Report: docs/Sem_4/final/FINAL_REPORT_VishalSingh_2020AA05641.pdf

Vishal Singh · 2020AA05641 · BITS Pilani (WILP) · Walmart Global Tech, Bengaluru